R13

Code No: 111AF

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year Examinations, December - 2017
COMPUTER PROGRAMMING
(Common to CE, EEE, ME, ECE, CSE, EIE, IT, MCT, MMT, AE, AME, MIE, PTM, CEE, AGE)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a)	Differentiate between compiler and interpreter.	[2]
b)	What is an algorithm? What is its role in program development?	[3]
c)	How many values can be returned by a function?	[2]
d)	Why is function known as a derived data type?	[3]
e)	What is dereference type?	[2]
f)	List the applications of pointers.	[3]
g)	What is the use of typedef?	[2]
h)	What is the difference between binary file and text file?	[3]
i)	What are the operations on stack?	[2]
j)	Compare array with linked list.	[3]

PART-B

(50 Marks)

2.a) Demonstrate the usage of relational operators.
b) Using while loop write a program to read 10 numbers from user and display their sum. [5+5]

OR

3.a) What is a data type? Discuss the four standard data types of C language.
b) Construct a flowchart to identify largest number among the three given numbers. [5+5]

4. Discuss the advantages of using functions in problem solving. With the help of suitable examples illustrate inter function communication. [10]

OR

5.a) How to declare and initialize a multi dimensional array?
b) Write a program to find sum of squares of an array element. [5+5]

6.a) Can one variable have many pointers? Write a program to justify your answer.
b) What operators are applicable to pointer variables? Give example expressions.
c) Write a function to swap two numbers using pointer variables to these numbers. [10]

OR

7.a) Write a program to check whether a given string is a palindrome or not.
b) Discuss the library functions for string to data conversion. [5+5]

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8.a) What is a structure? Define a structure and explain the two methods of accessing structure variables.

b) In which situation a union is preferred over union. [5+5]

OR

9.a) Write a program to transfer the contents of a file in to another file using file handling concepts.

b) How can errors be handled in file programs? [5+5]

10.a) Write an algorithm for binary search.

b) Perform insertion sort (step-by-step) and arrange the following number in ascending order. [5+5]

{23 45 25 67 89 37 12 90 32 19}

OR

11.a) Write a program to implement a queue using arrays.

b) What is an queue? Compare it with dequeue. [5+5]

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